

APPENDIX C: SERVICE LEVEL AGREEMENTS (SLAS), NETWORK KPIS & LIQUIDATED DAMAGES

5G FWA PROGRAM (5,000 SITES NATIONWIDE)

Tender Reference: RFP-TELTECH-FWA-2026-001

Project: Nationwide 5G FWA Project (Band n50, 60 MHz TDD)

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1. OBJECTIVE & GOVERNANCE PRINCIPLES

This Appendix defines the mandatory Service Level Agreements (SLAs), Key Performance Indicators (KPIs), measurement methodologies, reporting intervals, and financial remedies (Service Credits and Liquidated Damages) governing:

- 1. Turnkey Deployment & Commissioning Phases (Lots 1, 2, 5)
- 2. 5G Radio & Core Network Operational Performance (Lots 1, 3)
- 3. OSS/BSS Platform Reliability & Field Dispatch Operations (Lot 4)
- 4. Managed Services & First Line Maintenance Operations (Lot 6)

Performance shall be evaluated on a calendar monthly basis using the automated performance management OSS and audited ticketing records.

2. PROJECT ROLLOUT & DELIVERY SLAS

2.1 Milestone Delivery Schedule

The Contractor shall adhere strictly to the following turnkey delivery schedule for Phase 1 (1,000 sites):

- **Milestone 1 (TSSR & Nominal Engineering Approval):** Within **45 Calendar Days** of contract award for all 1,000 priority sites.
- **Milestone 2 (First 200 Sites Commercial On-Air):** Within **120 Calendar Days** of contract award.
- **Milestone 3 (Cumulative 500 Sites Commercial On-Air):** Within **210 Calendar Days** of contract award.

- **Milestone 4 (Cumulative 1,000 Sites Commercial On-Air & Integrated):** Within **365 Calendar Days** of contract award.

2.2 Delay Liquidated Damages (Rollout)

- If the Contractor fails to achieve Final Acceptance of a milestone by the scheduled completion date, the Operator shall deduct **Delay Liquidated Damages (DLD)** equal to **0.5% of the total milestone contract value per calendar week of delay**, up to a maximum aggregate cap of **10.0% of the total contract value**.
- Delays exceeding 60 calendar days beyond the milestone date shall entitle the Operator to terminate for default and draw down on the Contractor's Performance Bond.

3. RADIO ACCESS NETWORK (RAN) OPERATIONAL KPIS & SERVICE CREDITS

3.1 Core Radio KPIs (Calculated 24/7/365 Nationwide)

Radio Availability (%) = [(Total Operating Cell Hours – Total Unplanned Outage Cell Hours) / Total Opera

KPI Identifier	Metric Description	Measurement Methodology	Target Standard	Minimum Threshold	Service Credit Penalty (< Threshold)
KPI-RAN-01	Radio Network Availability	Monthly aggregate across all active cells	$\geq 99.95\%$	$\geq 99.70\%$	1.0% of monthly O&M fee per 0.1% shortfall
KPI-RAN-02	5G Session Setup Success (CSSR)	Initial RRC connection + 5G QoS flow setup	$\geq 99.60\%$	$\geq 99.00\%$	0.5% of monthly O&M fee per 0.2% shortfall
KPI-RAN-03	5G Session Drop Rate (CDDR)	Abnormal session drops during active data sessions	$\leq 0.15\%$	$\leq 0.50\%$	0.5% of monthly O&M fee per 0.1% excess
KPI-RAN-04	Peak Sector DL Throughput	Busy hour average across cell cluster	$\geq 450\text{ Mbps}$	$\geq 380\text{ Mbps}$	Mandatory RF re-tuning within 5 days
KPI-RAN-05	Cell-Edge DL User Speed	95th percentile stationary CPE speed	$\geq 50\text{ Mbps}$	$\geq 25\text{ Mbps}$	Mandatory antenna tilt optimization
KPI-RAN-06	Cell-Edge UL User Speed	95th percentile stationary CPE upload	$\geq 15\text{ Mbps}$	$\geq 5\text{ Mbps}$	Uplink power control re-configuration
KPI-RAN-07	Radio Round-Trip Latency (RTT)	CPE to local gNodeB/UPF edge	$\leq 18\text{ ms}$	$\leq 30\text{ ms}$	Audit packet buffering & backhaul
KPI-RAN-08	Packet Loss Rate (PLR)	User plane packet discard rate	$\leq 0.01\%$	$\leq 0.10\%$	Transport & radio QoS re-profiling

4. 5G CORE NETWORK & CLOUD PLATFORM KPIS

4.1 5GC High Availability Benchmarks

KPI Identifier	Metric Description	Measurement Target	Calculation Interval	Liquidated Remedy
KPI-5GC-01	Control Plane NF Availability (AMF/SMF/UDM)	$\geq 99.999\%$ ("Five Nines")	Monthly	\$10,000\text{ USD} per hour outage exceeding threshold
KPI-5GC-02	User Plane (UPF) Packet Availability	$\geq 99.999\%$	Monthly	\$10,000\text{ USD} per hour outage exceeding threshold
KPI-5GC-03	PDU Session Establishment Success Rate	$\geq 99.80\%$	24-hr Rolling	Root-cause analysis within 12 hours
KPI-5GC-04	CG-NAT Port Allocation Success Rate	$\geq 99.99\%$	Monthly	\$5,000\text{ USD} penalty for subscriber drop
KPI-5GC-05	Active-Active Datacenter Failover Time	$\leq 30\text{ Seconds}$	Semi-annual Drill	Penalty of 5% annual core AMC if drill fails
KPI-5GC-06	UPF Internal Packet Latency	$\leq 1.0\text{ ms}$	Continuous	Hardware NIC DPDK tuning

5. OSS/BSS & FIELD OPERATIONS SLAS

5.1 Commercial & Operational Velocity Metrics

KPI Identifier	Service Level Domain	Mandatory Target	Impact on Failure
KPI-BSS-01	Converged Charging System (CHF) Uptime	$\geq 99.99\%$	2.0% monthly BSS maintenance fee deduction
KPI-BSS-02	Top-Up Processing Speed (GCash / Maya)	$\leq 3.0\text{ Seconds}$	Mandatory optimization of API gateway queues
KPI-OSS-01	New Installation Dispatch Lead Time	$\leq 48\text{ Hours}$	$\geq 95\%$ of orders completed within 48h SLA
KPI-OSS-02	First-Time-Right (FTR) Installation Rate	$\geq 96.0\%$	Repeat truck rolls at Contractor expense
KPI-OSS-03	Churned CPE Hardware Recovery Rate	$\geq 70.0\%$	Contractor debited \$55\text{ USD} per unrecovered unit below 70%
KPI-OSS-04	Automated Work-Order Dispatch Engine Capacity	$\geq 3,500\text{ orders/day}$	Zero backlog accumulation during peak campaigns

6. MANAGED SERVICES & FIELD INCIDENT RESTORATION SLAS

6.1 Incident Classification & Mean Time to Restore (MTTR)

Incidents logged by the 24/7 NOC are classified into four severity tiers:

Severity Level	Definition / Operational Impact	Response SLA	MTTR SLA (Urban)	MTTR SLA (Rural)
Severity 1 (Critical)	Complete macro site outage; Core NF failure; >500 subscribers out of service; major backhaul	<= 15 Mins	<= 2.0 Hours	<= 4.0 Hours
Severity 2 (Major)	Single sector down; degraded throughput; loss of DC battery backup; generator fault	<= 30 Mins	<= 4.0 Hours	<= 8.0 Hours
Severity 3 (Minor)	Non-service affecting hardware fault; loss of redundancy (single PSU failed, SFP degraded)	<= 1 Hour	<= 12.0 Hours	<= 24.0 Hours
Severity 4 (Warning/Info)	Configuration inquiry, routine audit, planned preventative maintenance scheduling	<= 4 Hours	<= 48.0 Hours	<= 72.0 Hours

6.2 Service Credits for MTTR Breaches

For every hour (or fraction thereof) that a Severity 1 incident exceeds the mandated MTTR threshold, the Operator shall deduct:

- **Urban / Suburban Sites: 0.5% of monthly site Managed Services fee per hour** of prolonged outage.
- **Rural Sites: 0.25% of monthly site Managed Services fee per hour** of prolonged outage.
- Total monthly service credit deductions for Managed Services shall be capped at **20.0% of the total monthly Managed Services fee**.

7. MONTHLY REPORTING & GOVERNANCE CADENCE

1. **Daily Operational Flash Report:** Generated automatically at 06:00 PHT summarizing the previous 24 hours of network availability, outages, MTTR compliance, and work orders.
2. **Weekly Engineering Review:** Joint operations meeting reviewing open trouble tickets, cluster drive tests, and rollout milestone progress.
3. **Monthly Governance & SLA True-Up:** Formal executive session reviewing all KPI metrics against SLA thresholds, finalizing service credits, and approving contractor invoices.

(End of Appendix C — Service Level Agreements, Network KPIs & Liquidated Damages)